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B-26

Assignment: 6.3			
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Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals

Lab Objectives

- To explore AI-powered auto-completion features for core Python constructs such as classes, loops, and conditional statements.
- To analyze how AI tools suggest logic for object-oriented programming and control structures.
- To evaluate the correctness, readability, and completeness of AI-generated Python code.

Lab Outcomes (LOs)

After completing this lab, students will be able to:

- Use AI tools to generate and complete Python class definitions and methods.
- Understand and assess AI-suggested loop constructs for iterative tasks.
- Generate and evaluate conditional statements using AI-driven prompts.
- Critically analyze AI-assisted code for correctness, clarity, and efficiency.

Task Description #1: Classes (Student Class)

Scenario

You are developing a simple student information management module.

Task

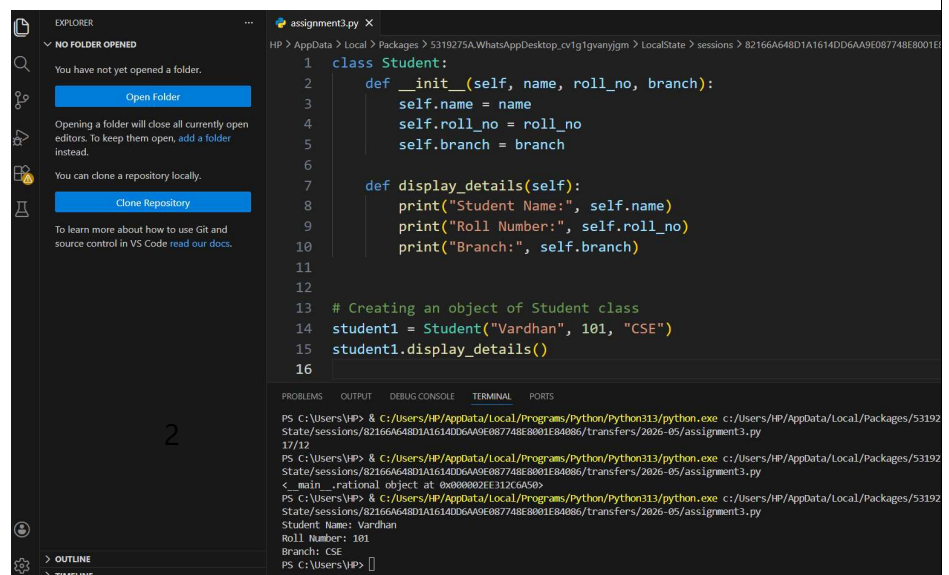
- Use an AI tool (GitHub Copilot / Cursor AI / Gemini) to complete a Student class.
- The class should include attributes such as name, roll number, and branch.
- Add a method `display_details()` to print student information.
- Execute the code and verify the output.
- Analyze the code generated by the AI tool for correctness and clarity.

Expected Output #1

- A Python class with a constructor (`__init__`) and a `display_details()` method.
- Sample object creation and output displayed on the console.
- Brief analysis of AI-generated code.

Prompt :

 Write a Python class named Student with attributes: name, roll_number, and branch. Include a method display_details that prints the student's information in a clear format. After defining the class, create an object of Student, assign sample values, and call display_details() to verify the output.



The screenshot displays the Visual Studio Code interface. On the left, the Explorer pane shows a project with no folders opened. The main editor area displays a file named `assignment3.py` containing the following Python code:

```
1 class Student:
2     def __init__(self, name, roll_no, branch):
3         self.name = name
4         self.roll_no = roll_no
5         self.branch = branch
6
7     def display_details(self):
8         print("Student Name:", self.name)
9         print("Roll Number:", self.roll_no)
10        print("Branch:", self.branch)
11
12
13 # Creating an object of Student class
14 student1 = Student("Vardhan", 101, "CSE")
15 student1.display_details()
16
```

Below the code editor, the TERMINAL pane shows the execution output:

```
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python313/python.exe c:/Users/HP/AppData/Local/Packages/5319275A/WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/82166A648D1A1614DD6AA9E087748E8001E84086/transfers/2026-05/assignment3.py
17/12
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python313/python.exe c:/Users/HP/AppData/Local/Packages/5319275A/WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/82166A648D1A1614DD6AA9E087748E8001E84086/transfers/2026-05/assignment3.py
<__main__.rational object at 0x00000000E312C6A0>
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python313/python.exe c:/Users/HP/AppData/Local/Packages/5319275A/WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/82166A648D1A1614DD6AA9E087748E8001E84086/transfers/2026-05/assignment3.py
Student Name: Vardhan
Roll Number: 101
Branch: CSE
PS C:\Users\HP>
```

Output :